

University of Nebraska-Lincoln

School of Computing

CSCE 473/873: Computer Vision
Fall 2026

Assignment 2: Image Filtering

The goal of this assignment is to write an image filtering function and use it to create hybrid images using a simplified version of the SIGGRAPH 2006 paper by Oliva, Torralba, and Schyns. Hybrid images are static images that change in interpretation as a function of the viewing distance. The basic idea is that high frequency tends to dominate perception when it is available, but, at a distance, only the low frequency (smooth) part of the signal can be seen. By blending the high frequency portion of one image with the low-frequency portion of another, you get a hybrid image that leads to different interpretations at different distances.

You must complete this assignment with the partner assigned to you.

Homework

This project is intended to familiarize you with MATLAB and image filtering. Once you have created an image filtering function, it is relatively straightforward to construct hybrid images. If you don't already know MATLAB, you will find this [tutorial on MATLAB](#) helpful.

Image Filtering. Image filtering (or convolution) is a fundamental image processing tool. MATLAB has numerous built-in and efficient functions to perform image filtering, but you will be writing your own such function from scratch for this assignment. More specifically, you will implement `my_imfilter()` which imitates the default behavior of the built-in `imfilter()` function. As specified in `my_imfilter.m`, your filtering algorithm must (1) support grayscale and color images, (2) support arbitrary shaped filters, as long as both dimensions are odd (e.g., 7×9 filters but not 4×5 filters), (3) pad the input image with zeros or reflected image content, and (4) return a filtered image which is the same resolution as the input image. We have provided a script, `test_filtering.m`, to help you debug your image filtering algorithm.

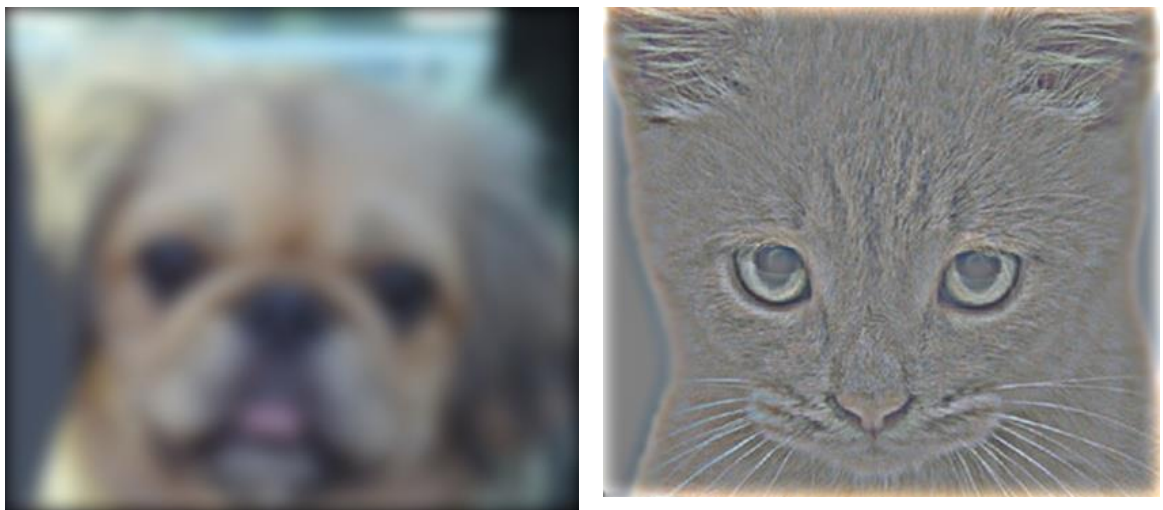
Hybrid Images. A hybrid image is the sum of a low-pass filtered version of the one image and a high-pass filtered version of a second image. There is a free parameter, which can be tuned for each image pair, which controls *how much* high frequency to

remove from the first image and how much low frequency to leave in the second image. This is called the "cutoff-frequency." In the paper, it is suggested to use two cutoff frequencies (one tuned for each image), and you are free to try that, as well. In the starter code, the cutoff frequency is controlled by changing the standard deviation of the Gaussian filter used in constructing the hybrid images.

We provide you with five (5) pairs of aligned images that can be merged reasonably well into hybrid images. The alignment is important because it affects the perceptual grouping (read the paper for details). We encourage you to create additional examples (e.g., change of expression, morph between different objects, change over time, etc.). For the example shown at the top of the page, the two original images look like this:



The low-pass (blurred) and high-pass versions of these images look like this:



The high frequency image is actually zero-mean with negative values. So, it is visualized by adding 0.5. In the resulting visualization, bright values are positive, and dark values are negative.

Adding the high and low frequencies together gives you the image at the top of this page. If you're having trouble seeing the multiple interpretations of the image, a useful way to visualize the effect is by progressively downsampling the hybrid image as is done below:



The starter code provides a function `vis_hybrid_image.m` to save and display such visualizations.

Potentially useful MATLAB functions: `fspecial()` and the operators in the [MATLAB tutorial](#) which make it efficient to cut out image subwindows and do the convolution (dot product) between them. `padarray()`.

Forbidden functions you can use for testing, but not in your final

code: `imfilter()`, `filter2()`, `conv2()`, `nlfilter()`, `colfilt()`.

Report

Summarize your work in a report. The report must be typewritten, formatted, and must be divided into sections corresponding to each task described above. In particular, you will describe your algorithm and any decisions you made to write your algorithm in a particular way. Then you will show and discuss the results of your algorithm. You must show the results of your filtering algorithm (the test script saves such images already). For extra credit, show some of the intermediate images in the hybrid image pipeline (e.g., the low and high frequency images, which the starter code already saves for you).

Also, discuss anything extra you did. Feel free to add any other information you feel is relevant. **The report must be in pdf format and no more than four (4) pages.**

Handin

Put all your files in a folder called HW2, compress it, and submit the compressed folder on Canvas. This is very important, as you will lose points if you do not follow instructions. Every instruction you don't follow, you will lose 5 points. The folder you hand in must contain the following:

- README - text file containing anything about the project that you want to tell the TA
- code/ - directory containing **all** your code for this assignment
- report/ - directory containing the report in pdf format

Due Date

The assignment is due on September 24 and is worth 65 points. The extra credit work is worth 5 additional points.

Rubric

- +30 pts: Working implementation of image filtering in `my_imfilter.m`
- +20 pts: Working hybrid image generation
- +15 pts: Report (must have several examples)
- -5*n pts: Lose 5 points for every time you do not follow the instructions for the submission of the assignment on Canvas

Credits

Assignment adapted by Ashok Samal based on a similar assignment by James Hays.